

Karen Williams

Data Science / ML / AI — West Franciscobury, Australia

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Profile

Senior data science / ml / ai engineer with 5 years of experience architecting and delivering enterprise solutions. Expert in Go, R, LightGBM with demonstrated leadership in cross-functional teams.

Experience

Research Scientist at Amazon 2025-09 – Present

Implemented A/B testing framework using Scala, running 10000+ experiments
Developed ML models using Redshift, improving prediction accuracy by 23%
Created recommendation system using Scala, increasing user engagement by 55%
Developed data visualization tools using R, used by 100000+ analysts

Research Scientist at Uber 2021-06 – 2025-04

Designed data warehousing solution using Lambda Labs, enabling analytics on 500+ records
Created feature engineering pipeline using Pandas, reducing preprocessing time by 68%
Built ML model serving infrastructure using LightGBM, handling 50000+ predictions/day
Developed ML models using LightGBM, improving prediction accuracy by 33%

AI Product Engineer at TechFlow AI 2020-06 – 2021-04

Built end-to-end ML pipeline using Qdrant, processing 1000+ samples daily
Built ML model serving infrastructure using Qdrant, handling 10000+ predictions/day
Built real-time analytics dashboard using Vector Databases, used by 50000+ stakeholders

Education

M.Tech., Electrical Engineering 2018

University of Washington

Technical Skills

Go, R, LightGBM, OpenCV, Claude API, Neptune, NumPy, DVC, Spark, Dask, Weaviate, Snowflake, GCP Vertex AI, AWS SageMaker, Lambda Labs, RAG, Fine-tuning, Computer Vision, Prompt Engineering

Certifications

AWS Certified Machine Learning (Amazon Web Services, 2025)
Databricks Certified Data Engineer (Databricks, 2020)

Other

Reference: References available upon request